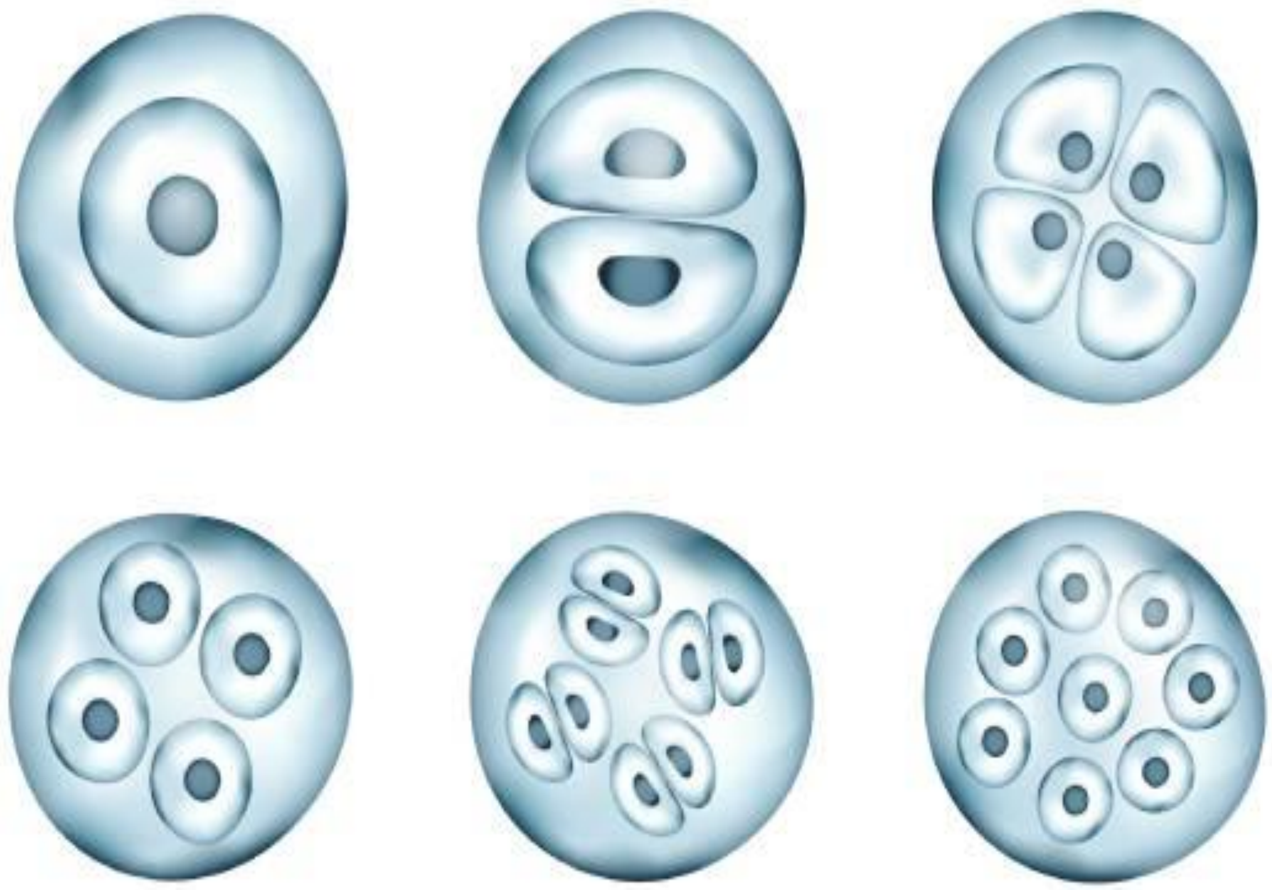

Introduction to Biology
BIO 101

Cell Division
Lecture 04

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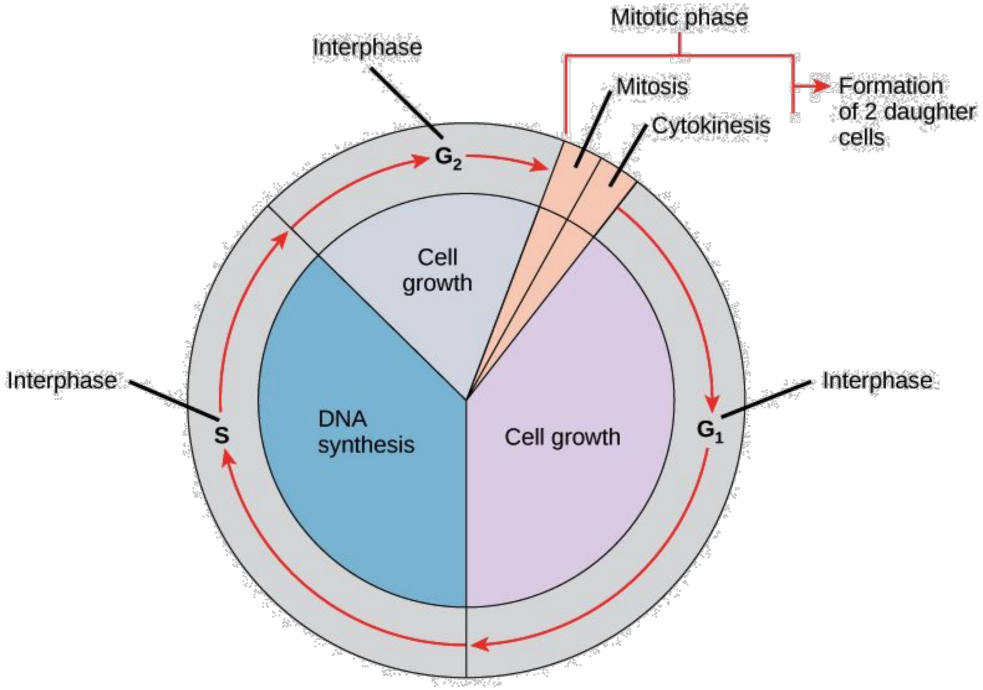
CELL CYCLE

The cell cycle is an ordered series of events involving cell growth and cell division that produces new daughter cells.

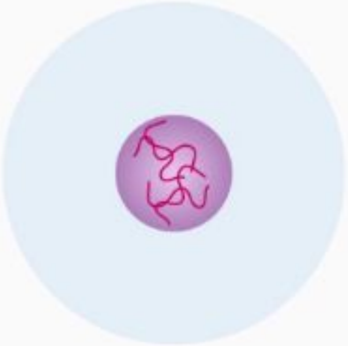


CELL DIVISION

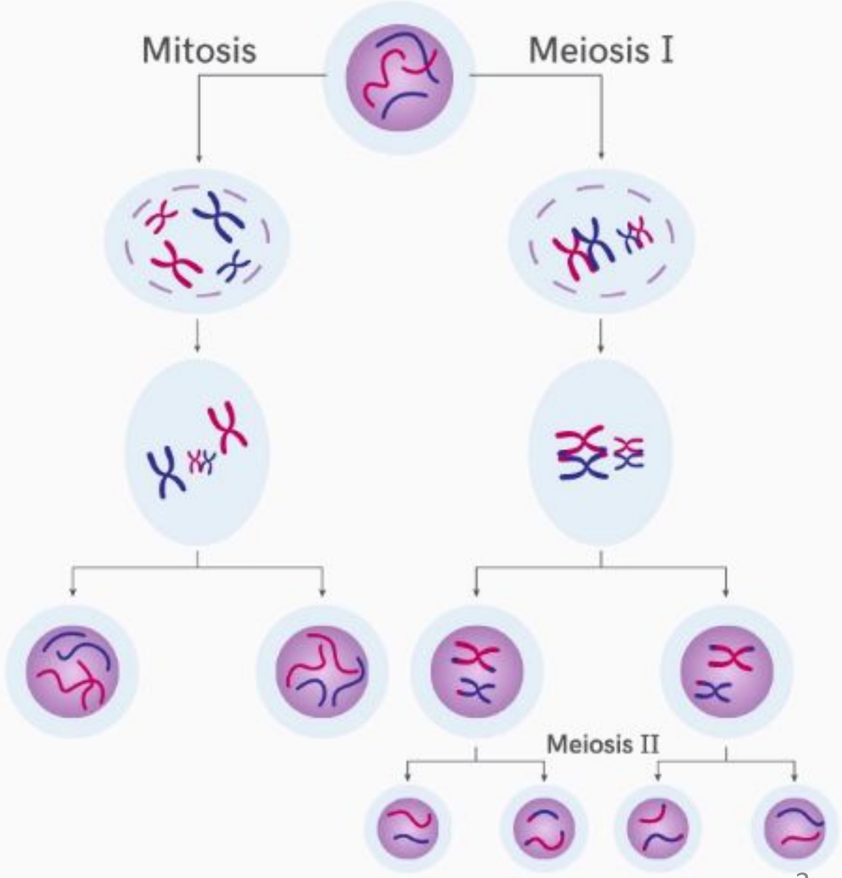
Cell division is the process by which a parent cell divides into two or more daughter cells. Cell division usually occurs as part of a larger cell cycle.

Cell Cycle

Interphase
(Preparatory Phase)
- Cell Growth
- DNA replication



M Phase
(Mitotic/Meiotic Phase)
- Cell Division



~ 24 hours to complete one cycle

Interphase → More than 90% of total time of Cell Cycle

M Phase → Less than 10% of total time of Cell Cycle

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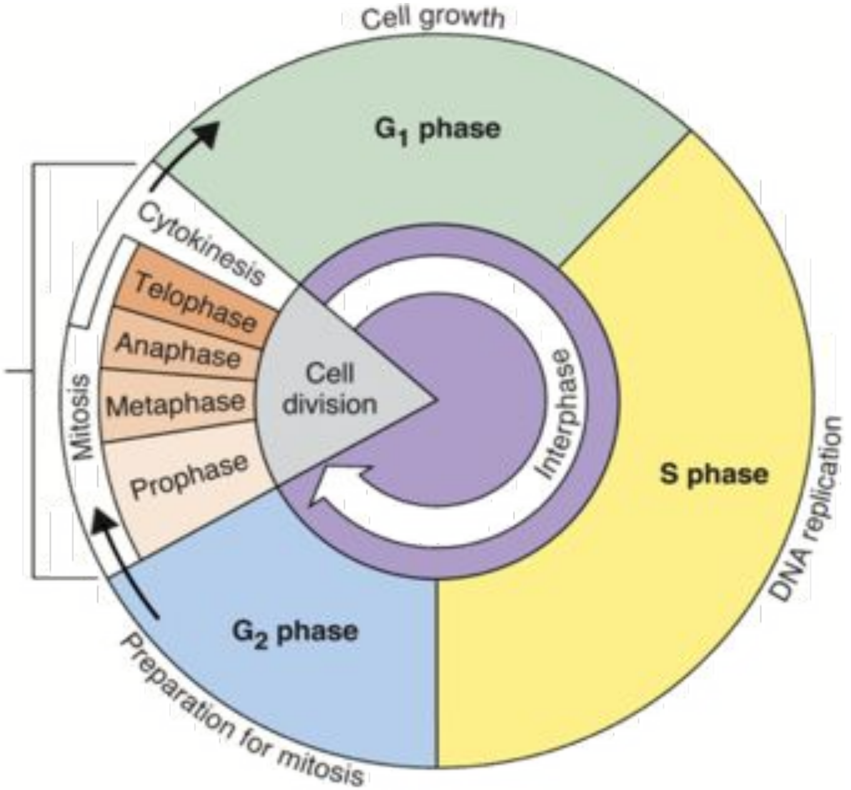
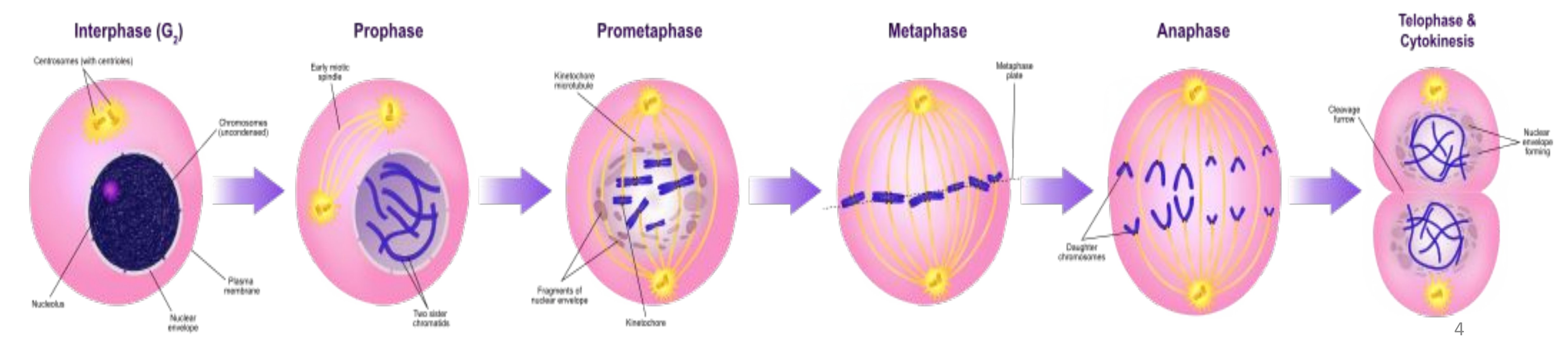
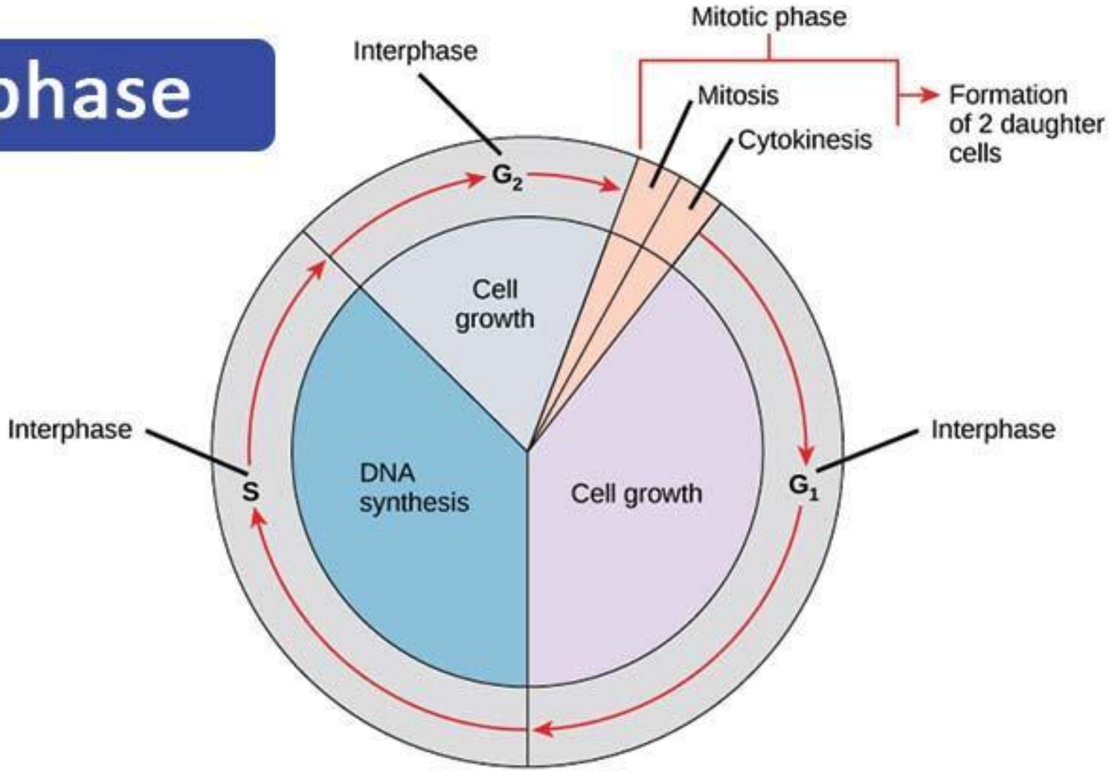




Interphase



- Usually, longest part of the cycle
- Cell increases in mass
- Number of cytoplasmic components doubles
- DNA is duplicated

Stages:

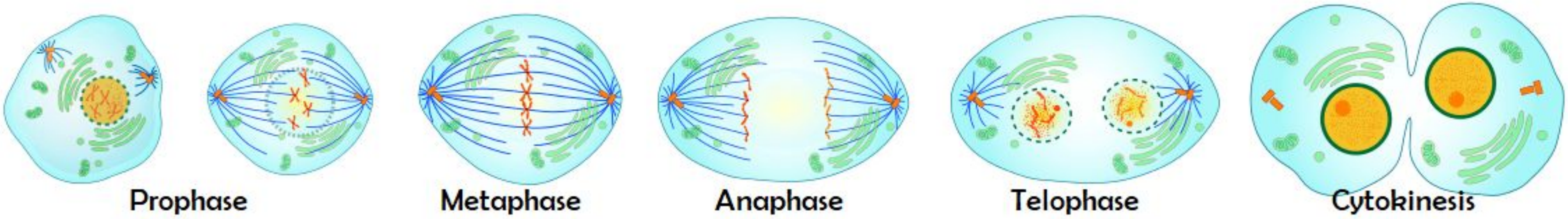
- **G1 :** Interval or gap after cell division – organelles copied, cells grow
- **S**: Time of DNA synthesis (replication) - Each DNA is copied
- **G2**: Interval or gap after DNA replication – cells grow more, reorganize contents

MITOSIS CELL DIVISION

Period of nuclear DNA division
Usually followed by cytoplasmic division

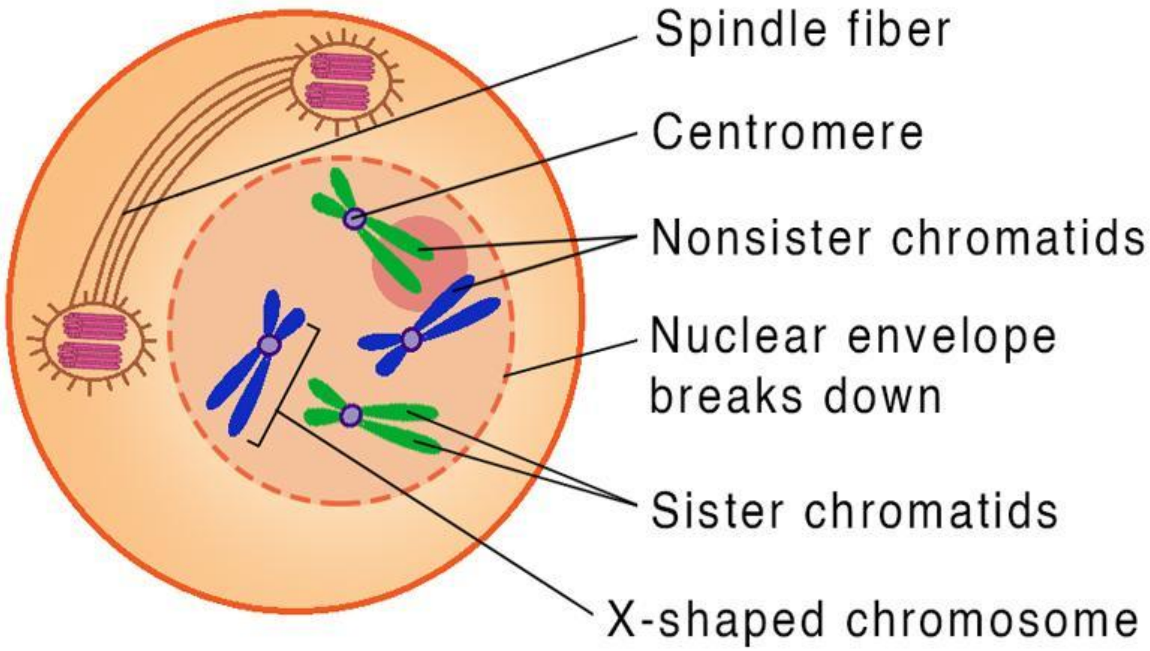
Four stages

Prophase
Metaphase
Anaphase
Telophase



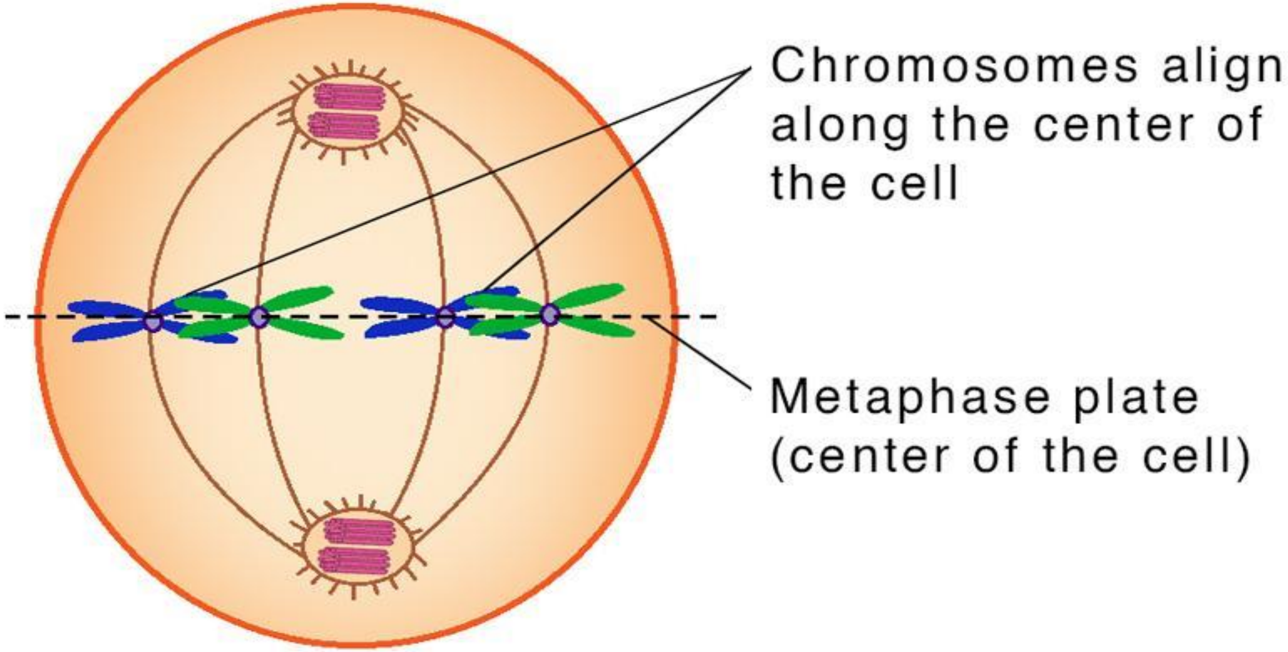
PROPHASE

1. Duplicated chromosomes begin to condense
2. New microtubules are assembled, mitotic spindle begins to form
3. One centriole pair (centrosome) is moved toward opposite pole of spindle
4. Nuclear envelope starts to break up
5. Spindle forms
6. Spindle microtubules become attached to the two sister chromatids of each chromosome



METAPHASE

1. All chromosomes are lined up at the spindle equator
2. Chromosomes are maximally condensed
3. Spindle checkpoint

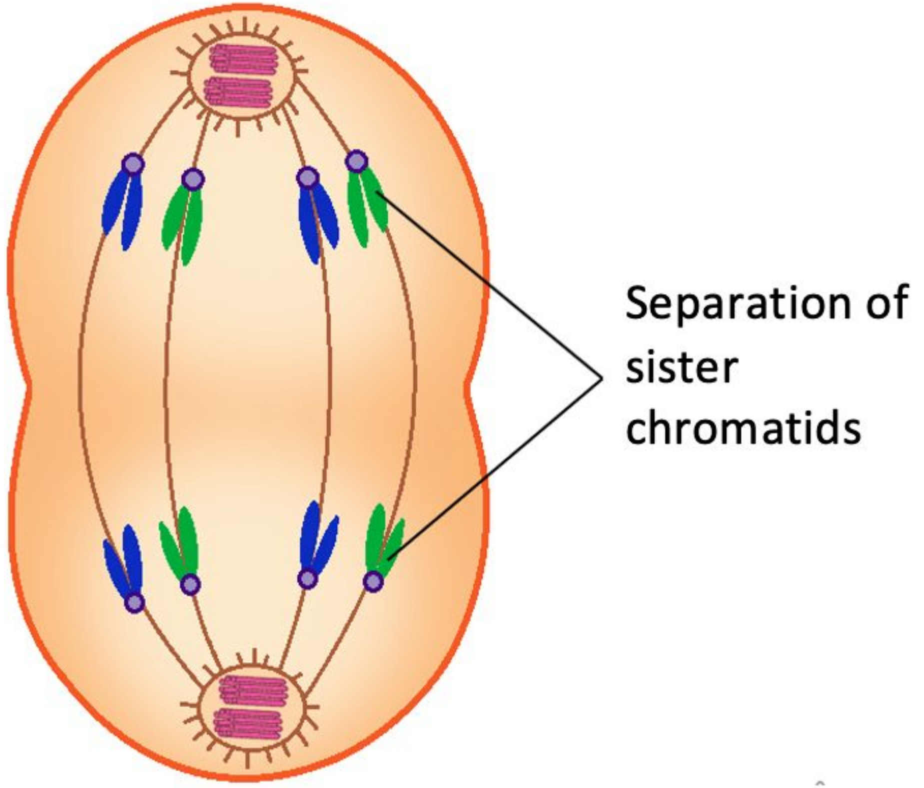


ANAPHASE

1. Sister chromatids of each chromosome are pulled apart towards poles due to contraction of spindle fiber

2. Once separated, each sister chromatid is a daughter chromosome

3. At the end of this phase, each pole contains a complete set of chromosomes.

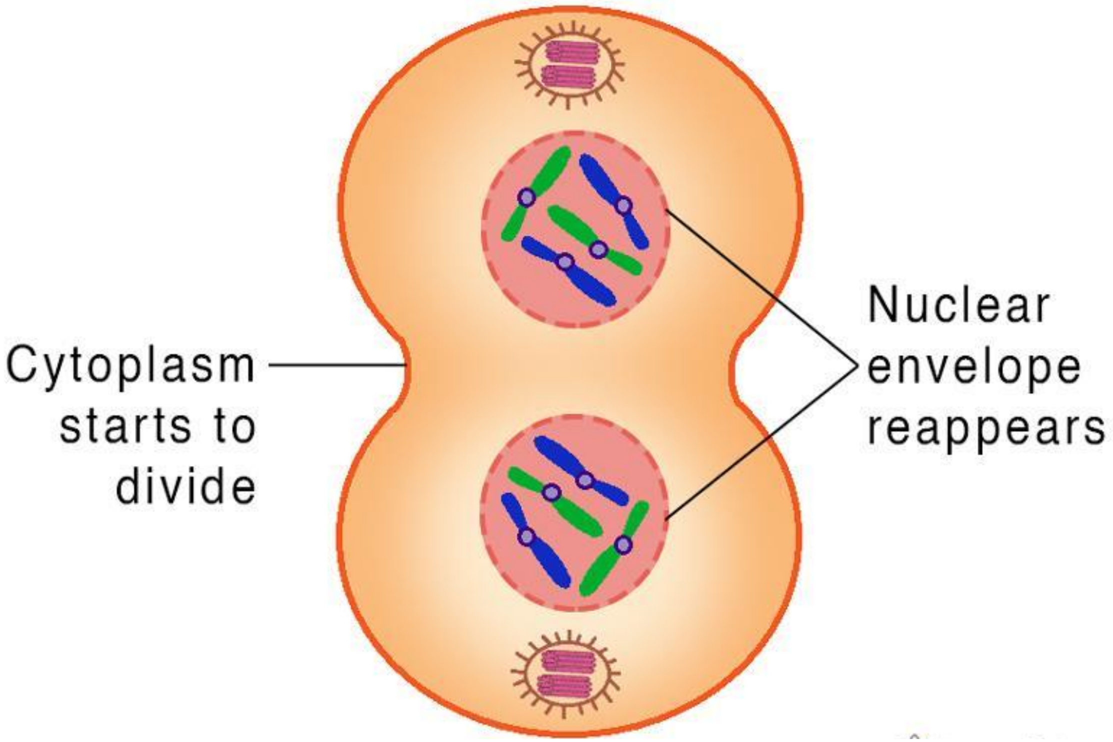


TELOPHASE

1. Chromosomes de-condense

2. Spindle disappears

3. Nuclear envelope appears, 2 new nuclei form for each set of chromosomes



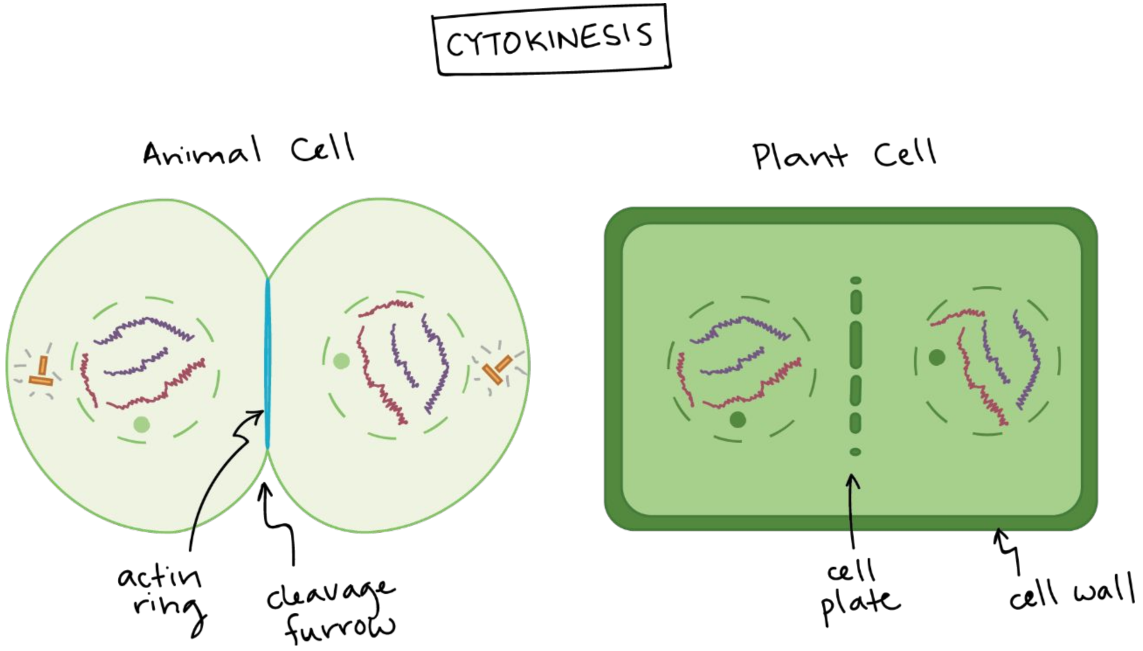
CYTOKINESIS

- Cytoplasmic Division

- Usually occurs between late anaphase and end of telophase

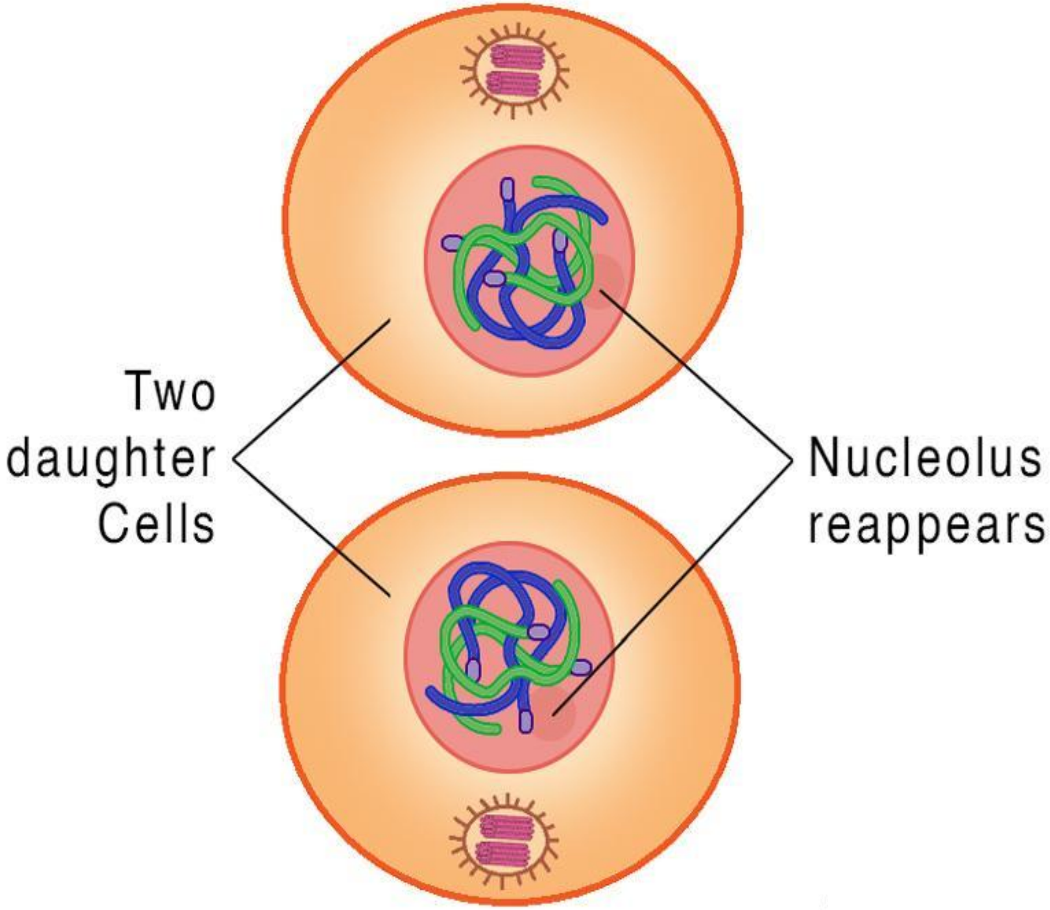
Two mechanisms

Through cleavage furrow (animals)
Through cell plate formation (plants)

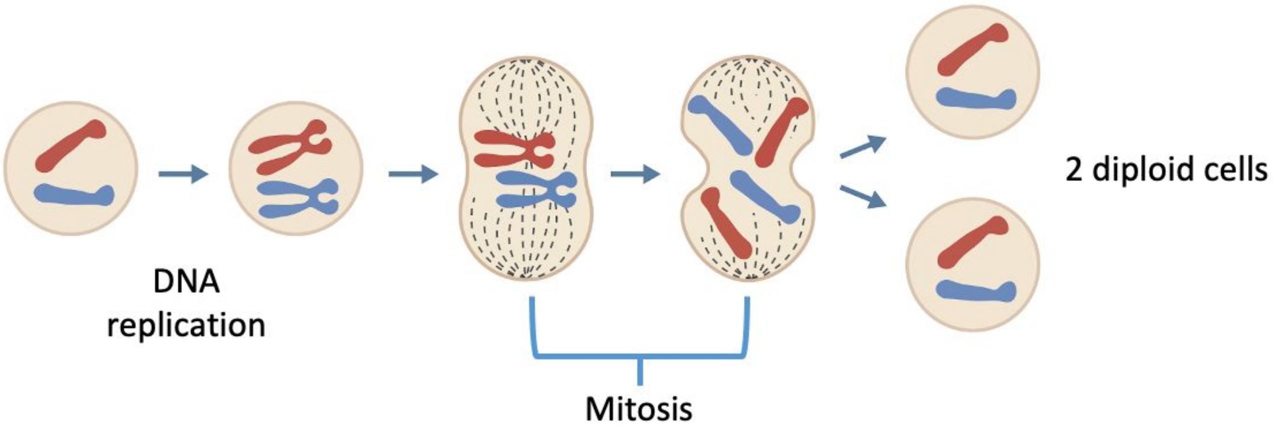


RESULTS OF MITOSIS

1. Two daughter nuclei
2. Each with same chromosome number as parent cell
3. Chromosomes in relaxed form



SIGNIFICANCE OF MITOSIS



1. Responsible for the growth and development of multicellular organisms. Facilitates the development of a single-cell zygote into a full-grown adult.

2. Hereditary material equally distributed in the daughter cells; no crossing over creates no variation

3. Helps in repairing damaged or replacement of older cells.

4. Some organisms undergo asexual reproduction by mitosis.

5. Malfunction leads to unwanted tumor or cancer.

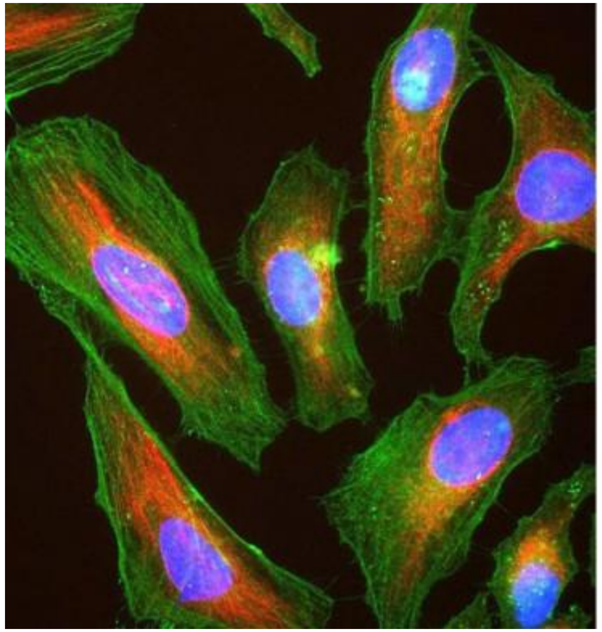
HELA CELLS

- Growing cells in culture allows researchers to investigate processes and test treatments without danger to patients.

- Most cells cannot be grown in culture.
- Line of human cancer cells that can be grown in culture.

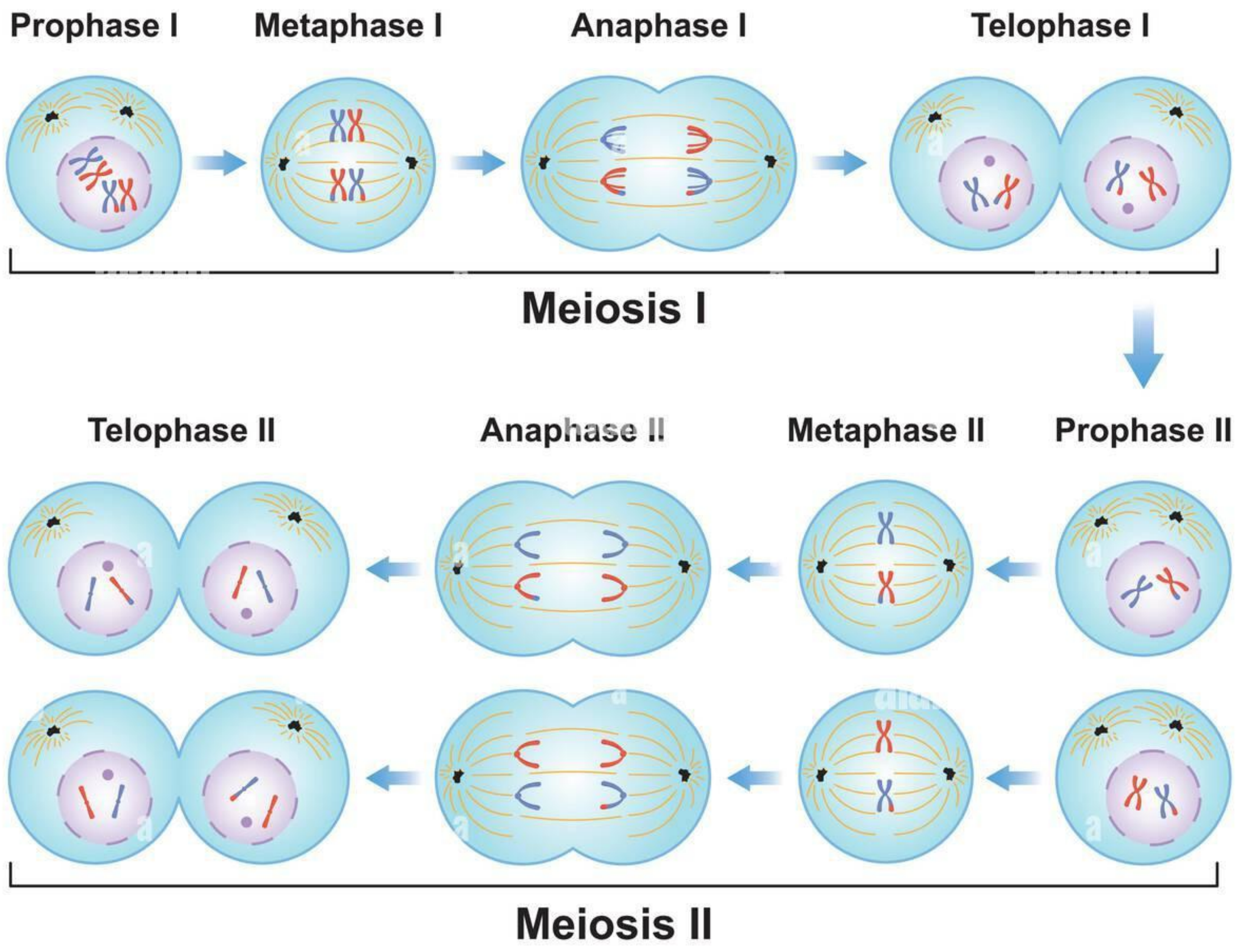
- Descendants of tumor cells from a woman named Henrietta Lacks.

- Lacks died at 31 in 1951, but her cells continue to live and divide in labs around the world.

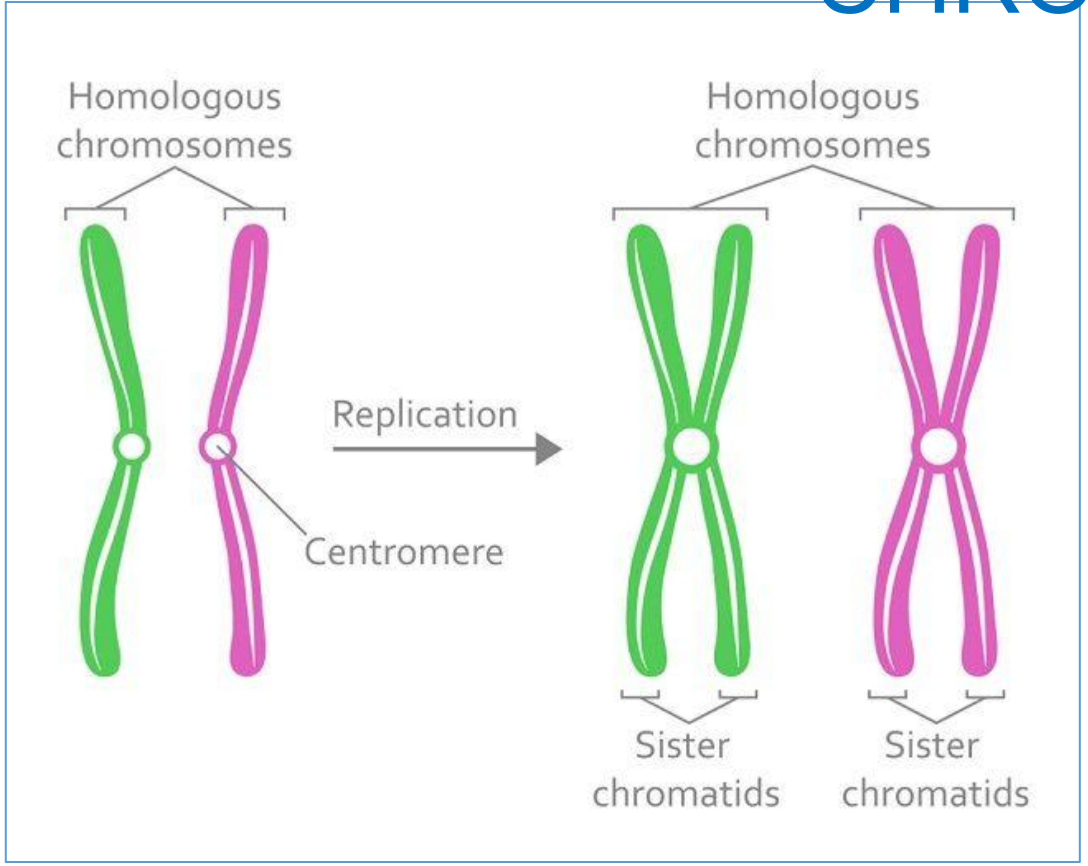


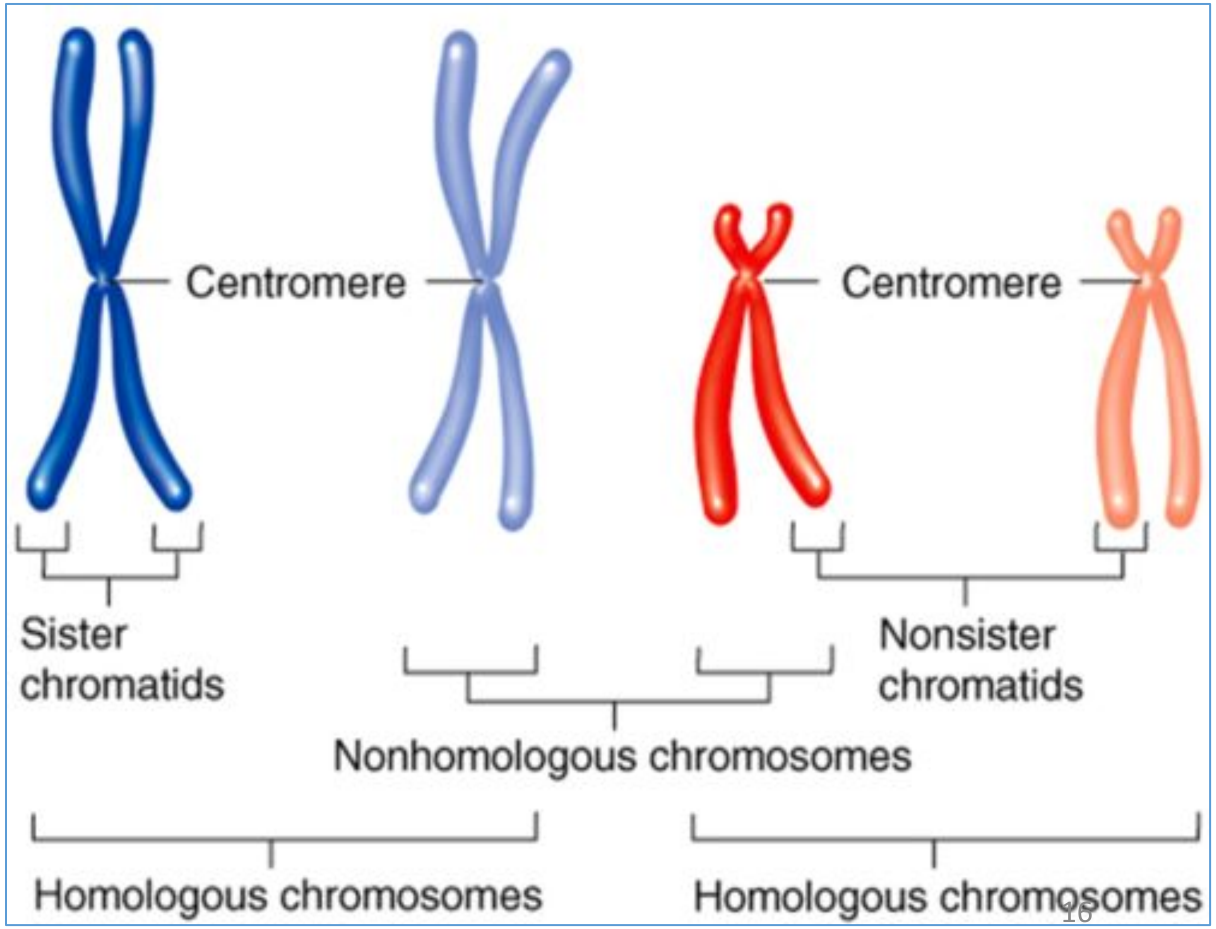
- These cells have been invaluable for medical research:
  - Used to create the polio vaccine
  - First human cells to go up in space to see the effect at zero gravity
  - Vital in cloning
  - In vitro fertilization
  - Gene mapping

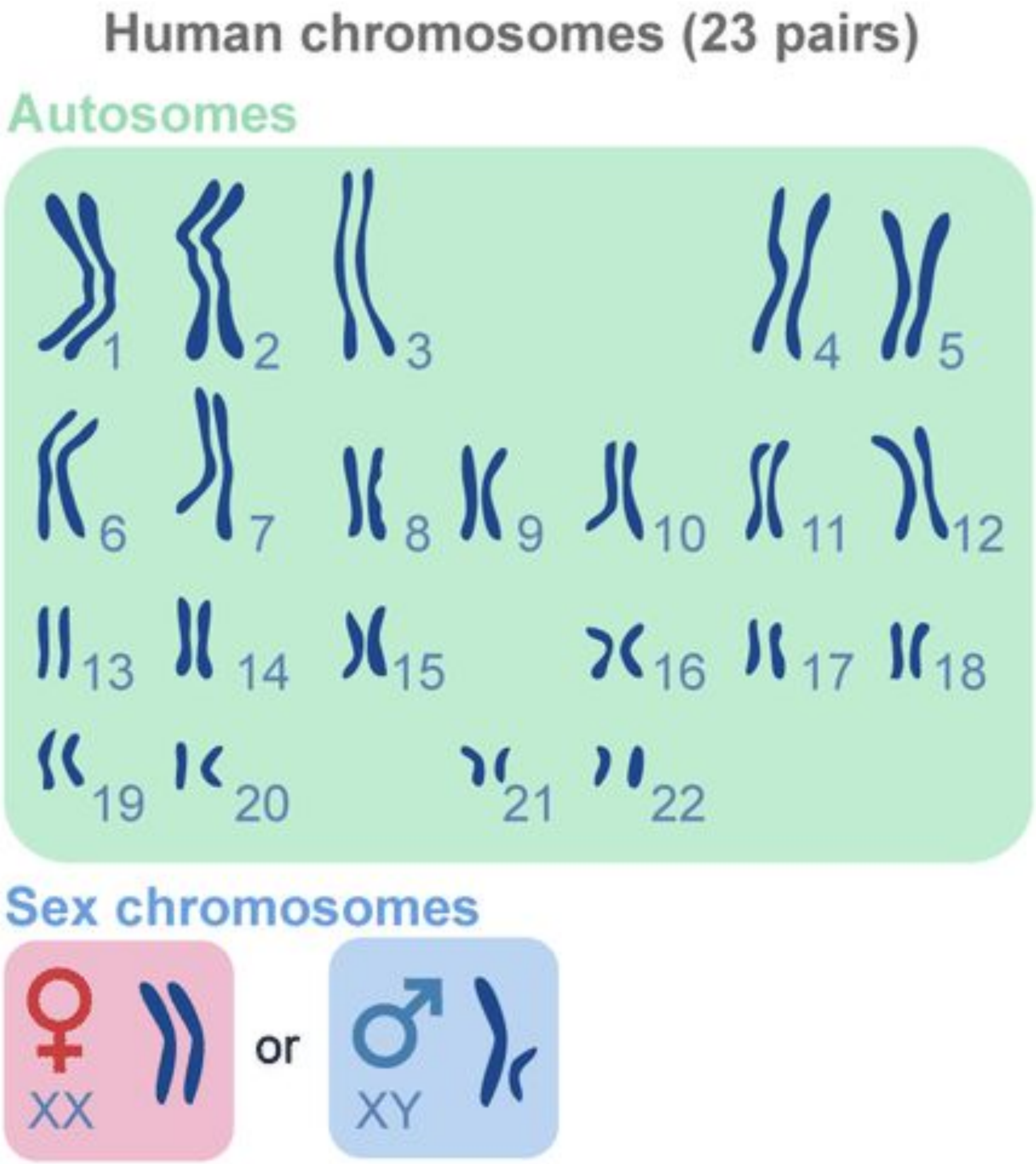
MEIOSIS CELL DIVISION



HOMOLOGOUS CHROMOSOMES

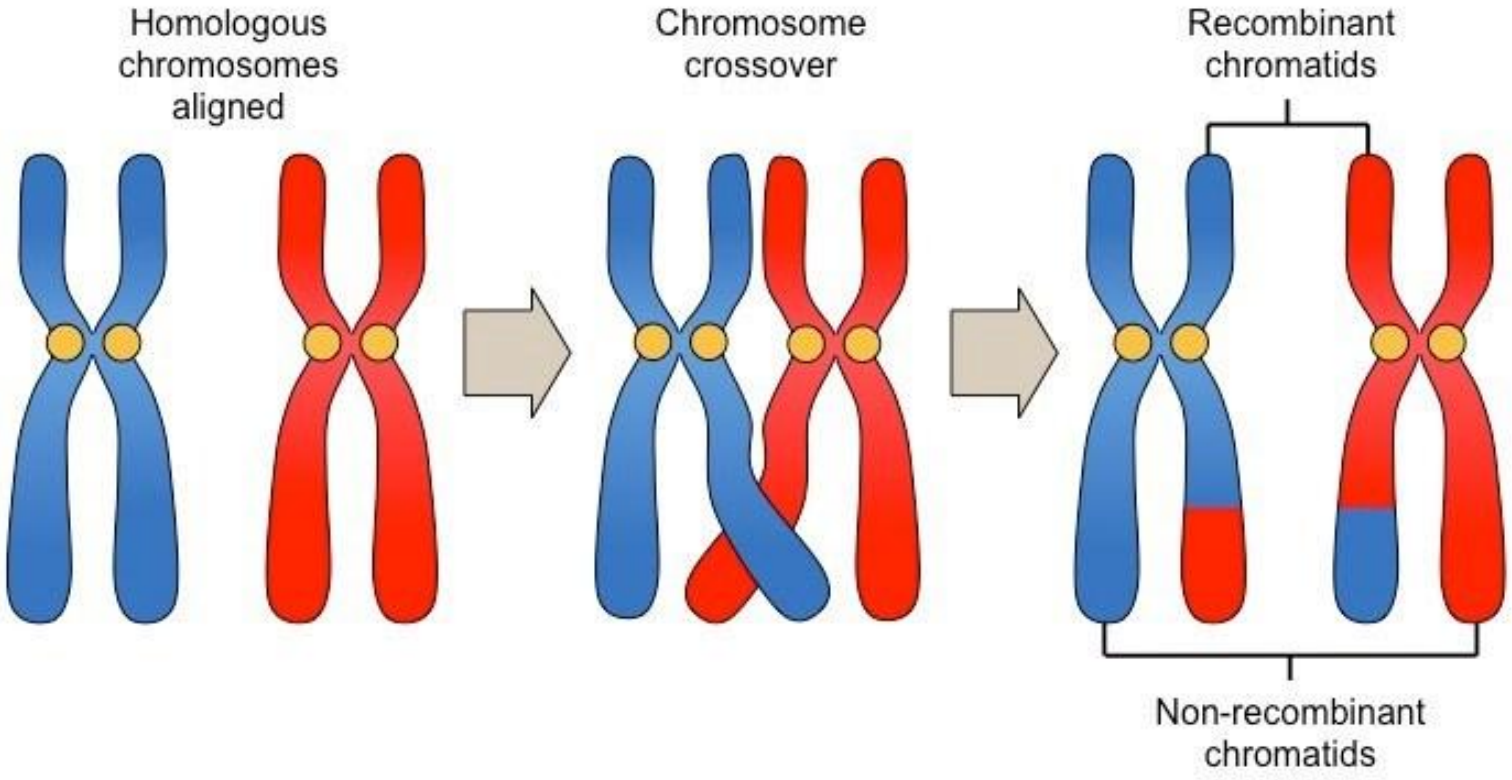




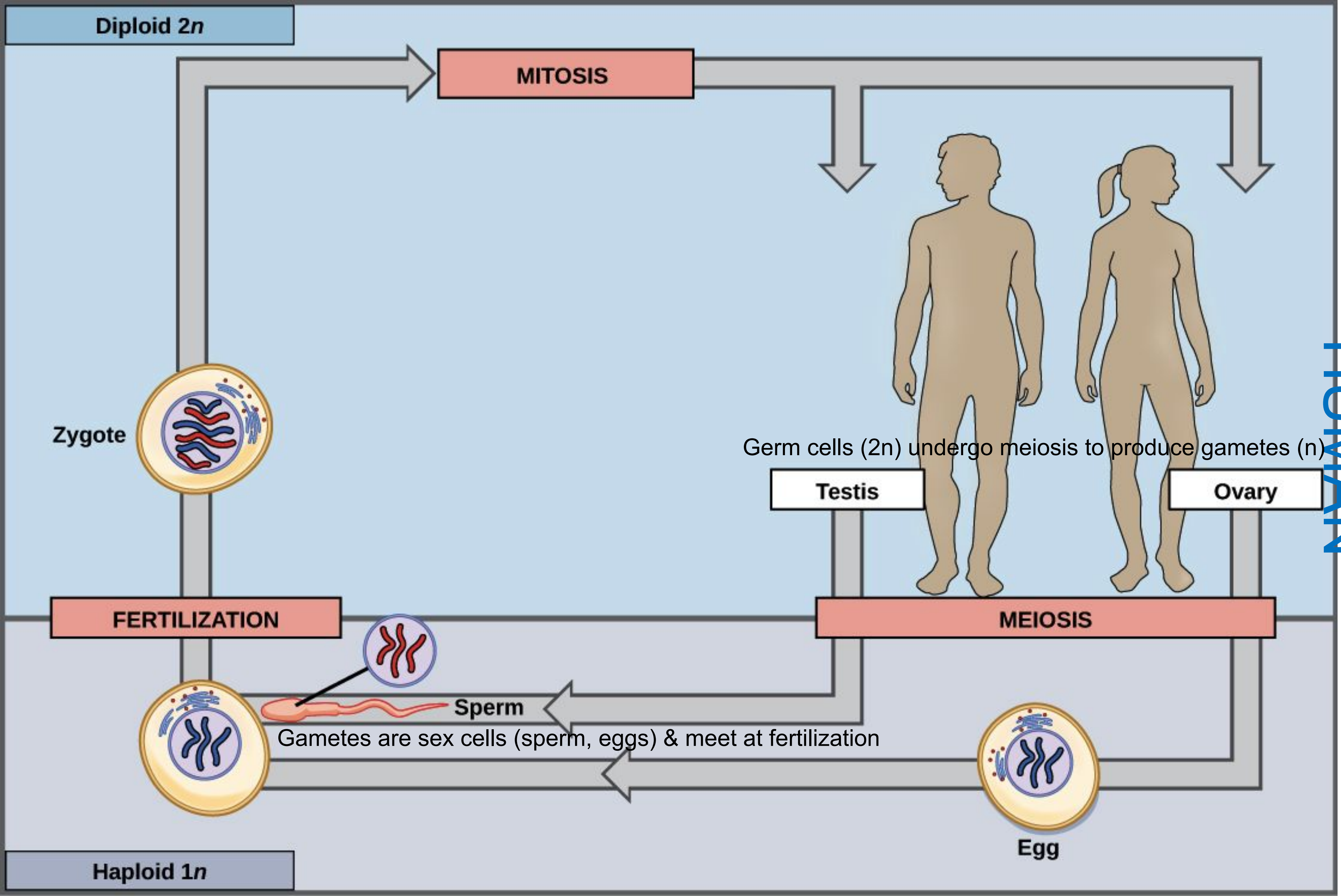

Human chromosomes (23 pairs)
Autosomes
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
Sex chromosomes
XX
or
XY

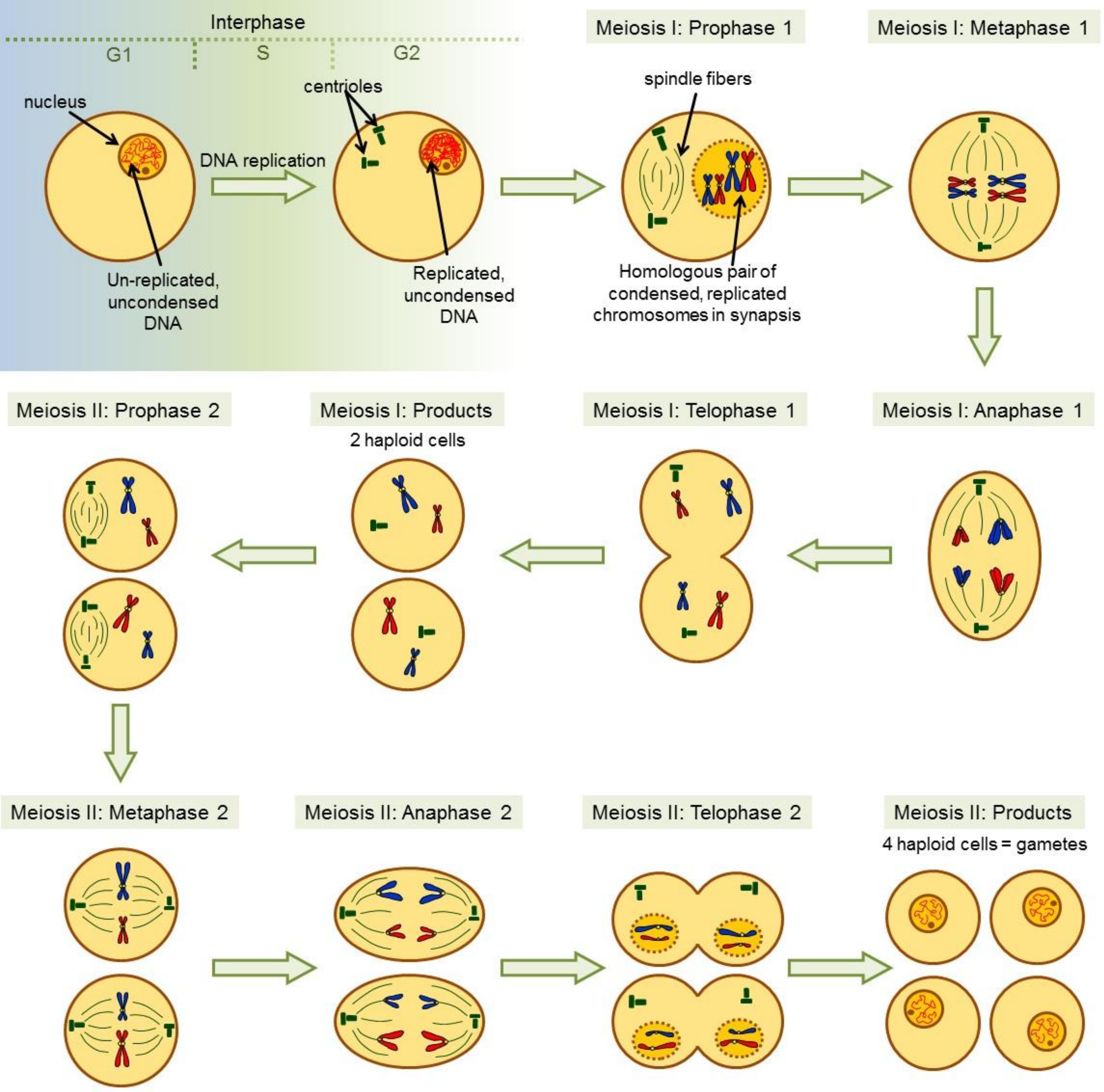
CROSSING OVER

- Each chromosome becomes zippered to its homologue
- All four chromatids are closely aligned
- Non-sister chromatids exchange segments
- After crossing over, each chromosome contains both maternal and paternal segments



M PHASES IN HUMAN





MEIOSIS

Two consecutive nuclear divisions

Meiosis I

Meiosis II

DNA is NOT duplicated between divisions

Four haploid nuclei are formed

Meiosis I

Prophase I

Metaphase I

Anaphase I

Telophase I

Meiosis II

Prophase II

Metaphase II

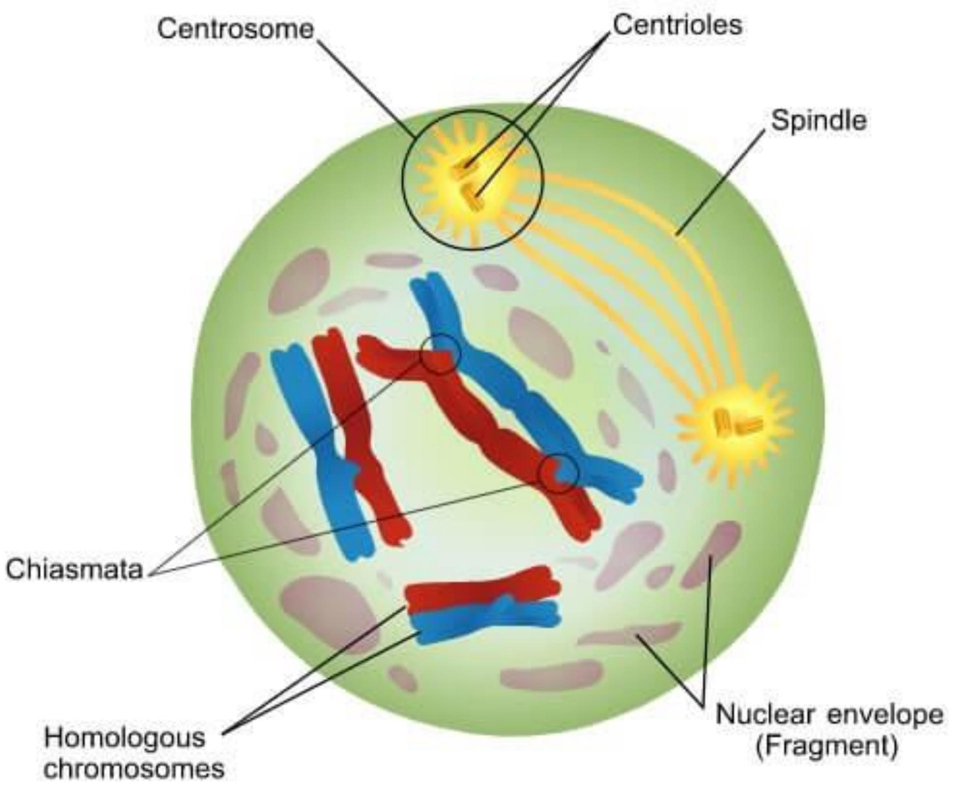
Anaphase II

Telophase II

MEIOSIS I

PROPHASE I

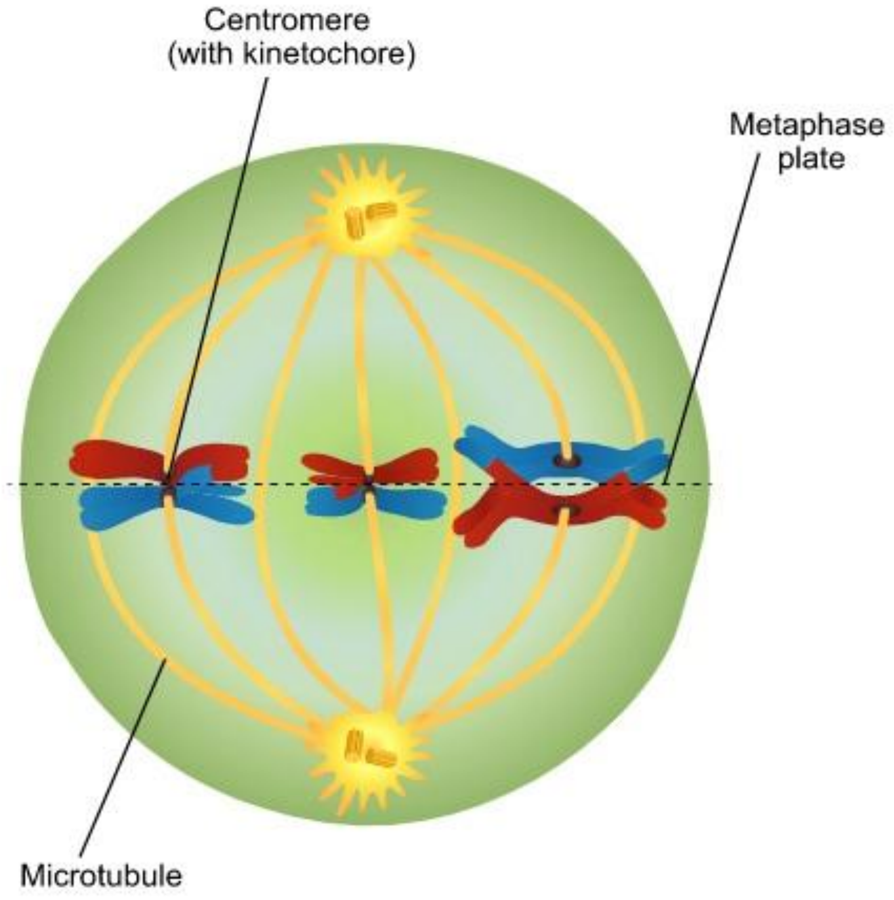
1. Each duplicated, condensed chromosome pairs with its homologue
2. Homologues swap segments: **Crossing over**
3. Nuclear envelope disappears
4. Each chromosome becomes attached to microtubules of newly forming spindle



MEIOSIS I

METAPHASE I

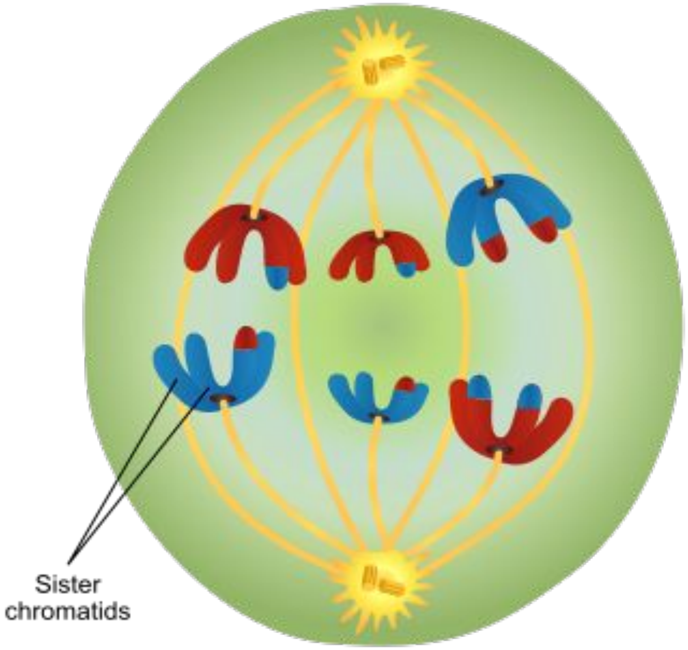
1. Chromosomes are pushed and pulled into the middle of cell
2. Sister chromatids of one homologue orient toward one pole, and those of other homologue toward opposite pole
3. The spindle is now fully formed



MEIOSIS I

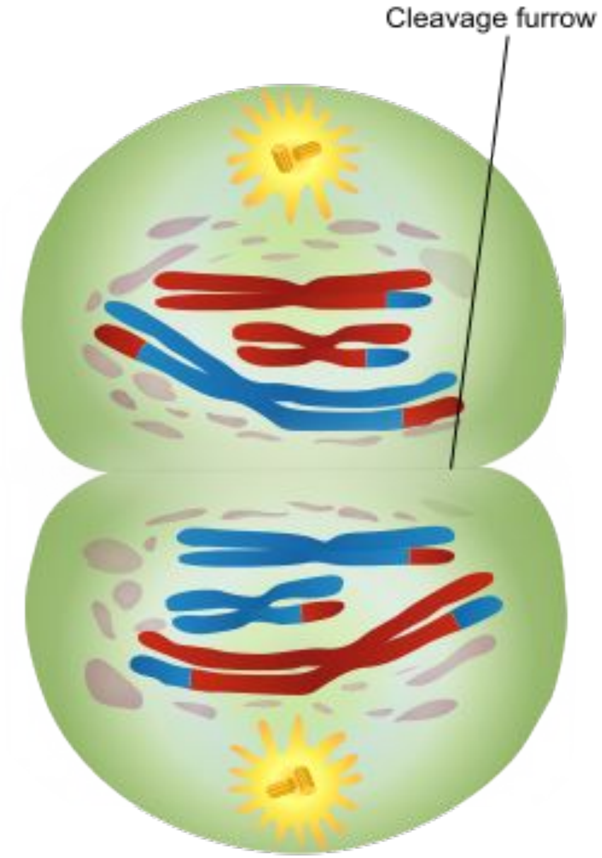
ANAPHASE I

1. **Homologous chromosomes** segregate from each other
2. The sister chromatids of each chromosome remain attached



TELOPHASE I

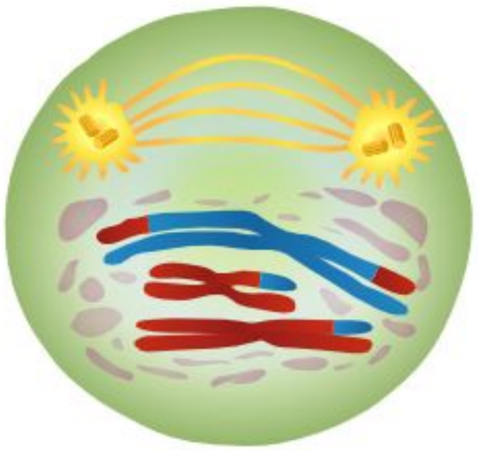
1. The chromosomes arrive at opposite poles
2. The cytoplasm divides
3. There are now **two haploid cells**
4. This completes Meiosis I



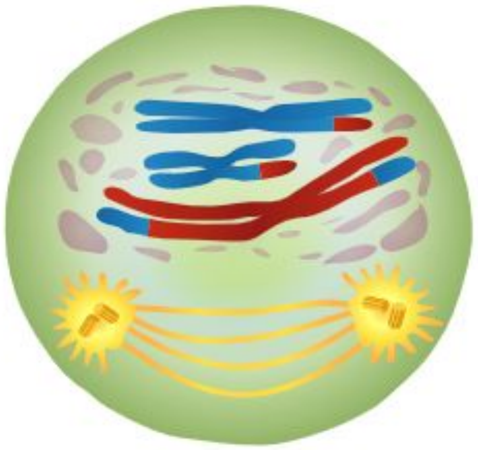
MEIOSIS II

PROPHASE II

1. Nuclear envelope degrades
2. A new spindle forms around the chromosome
3. Microtubules attach to the kinetochores of the duplicated chromosomes



A new spindle forms around the chromosomes.



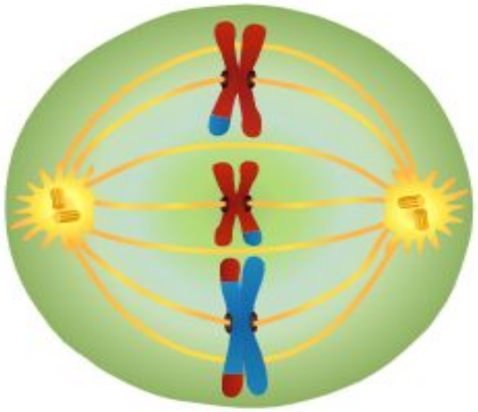
MEIOSIS II

METAPHASE II

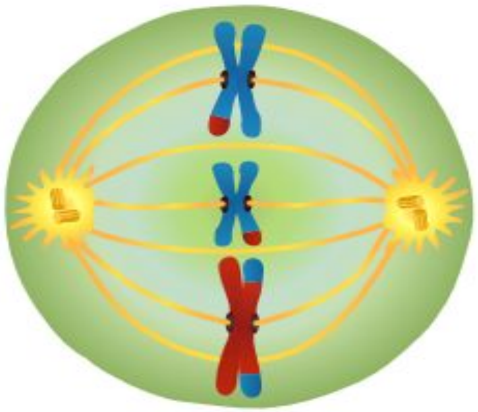
1. Chromosomes are pushed and pulled into the middle of cell

2. All of the duplicated chromosomes are lined up at the spindle equator, midway between the poles

3. Each sister chromatid of a chromosome would face opposite poles



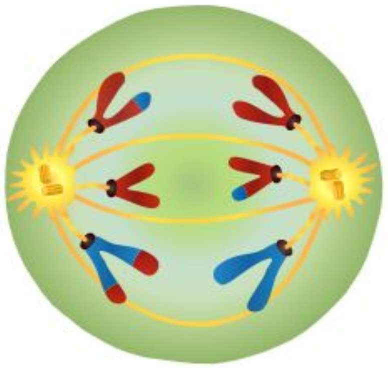
Metaphase II chromosomes line up at the equator.



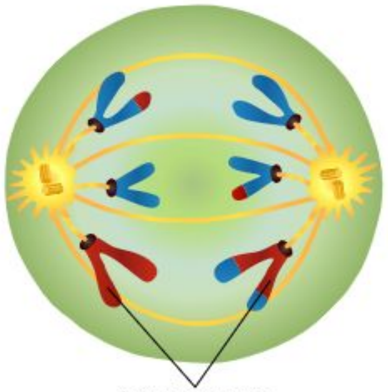
MEIOSIS II

ANAPHASE II

Sister chromatids separate to become independent chromosomes

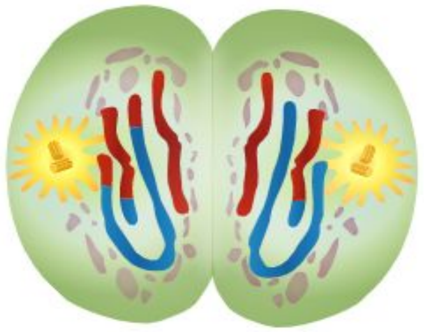


Centromeres divide. Chromatids move to the opposite poles of the cells.

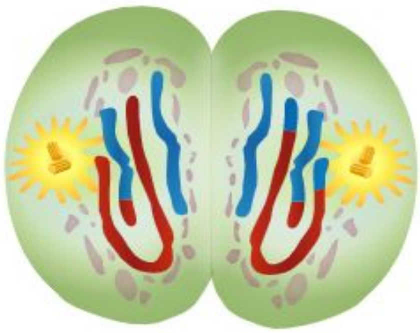


TELOPHASE II

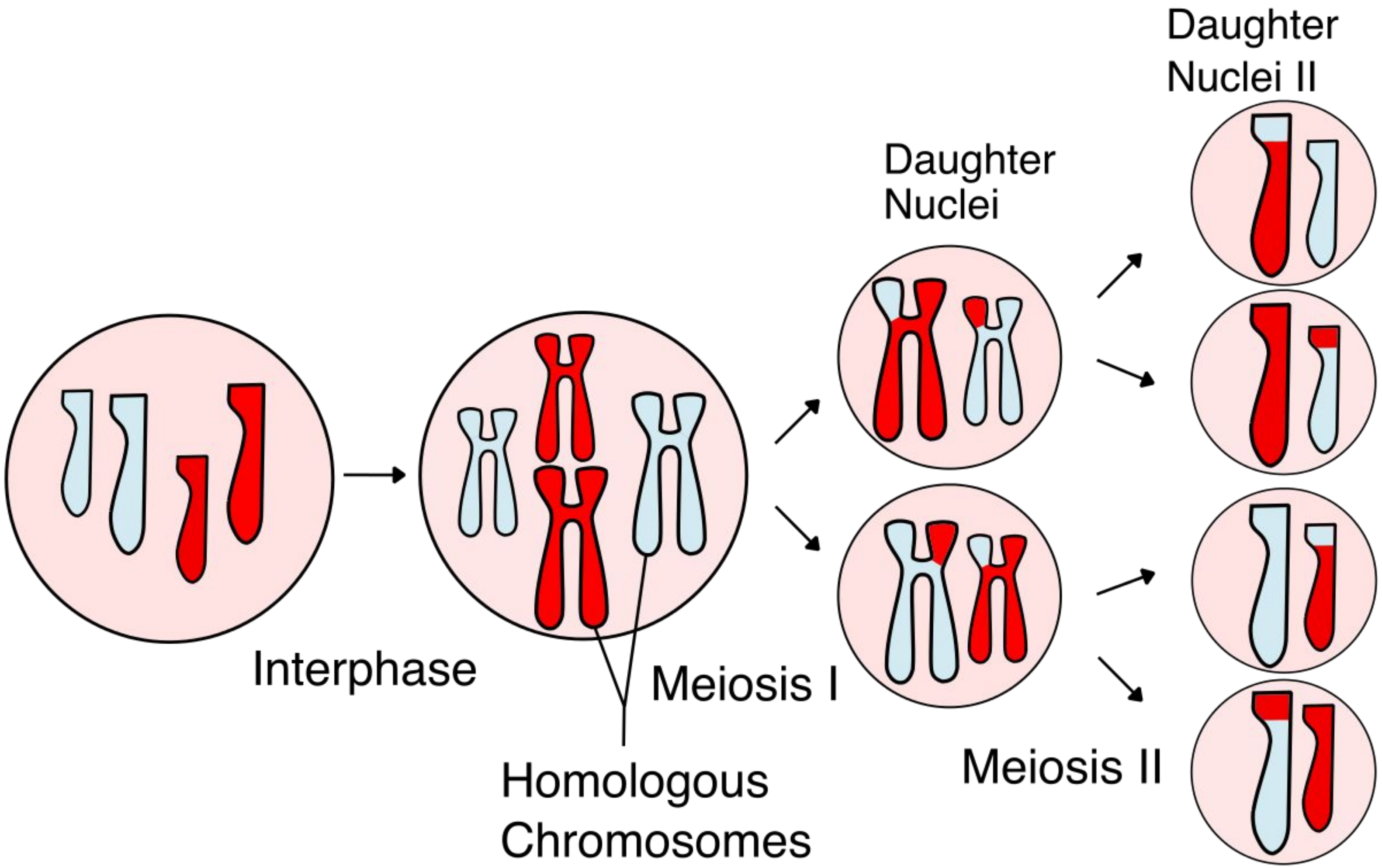
1. The chromosomes arrive at opposite ends of the cell
2. A nuclear envelope forms around each set of chromosomes & the cytoplasm divides
3. There are now four haploid cells



A nuclear envelope forms around each set of chromosomes. The cytoplasm divides.



RESULTS OF MEIOSIS



- Four haploid cells produced
- Differ from parent and one another: Crossing over and Random Assortment

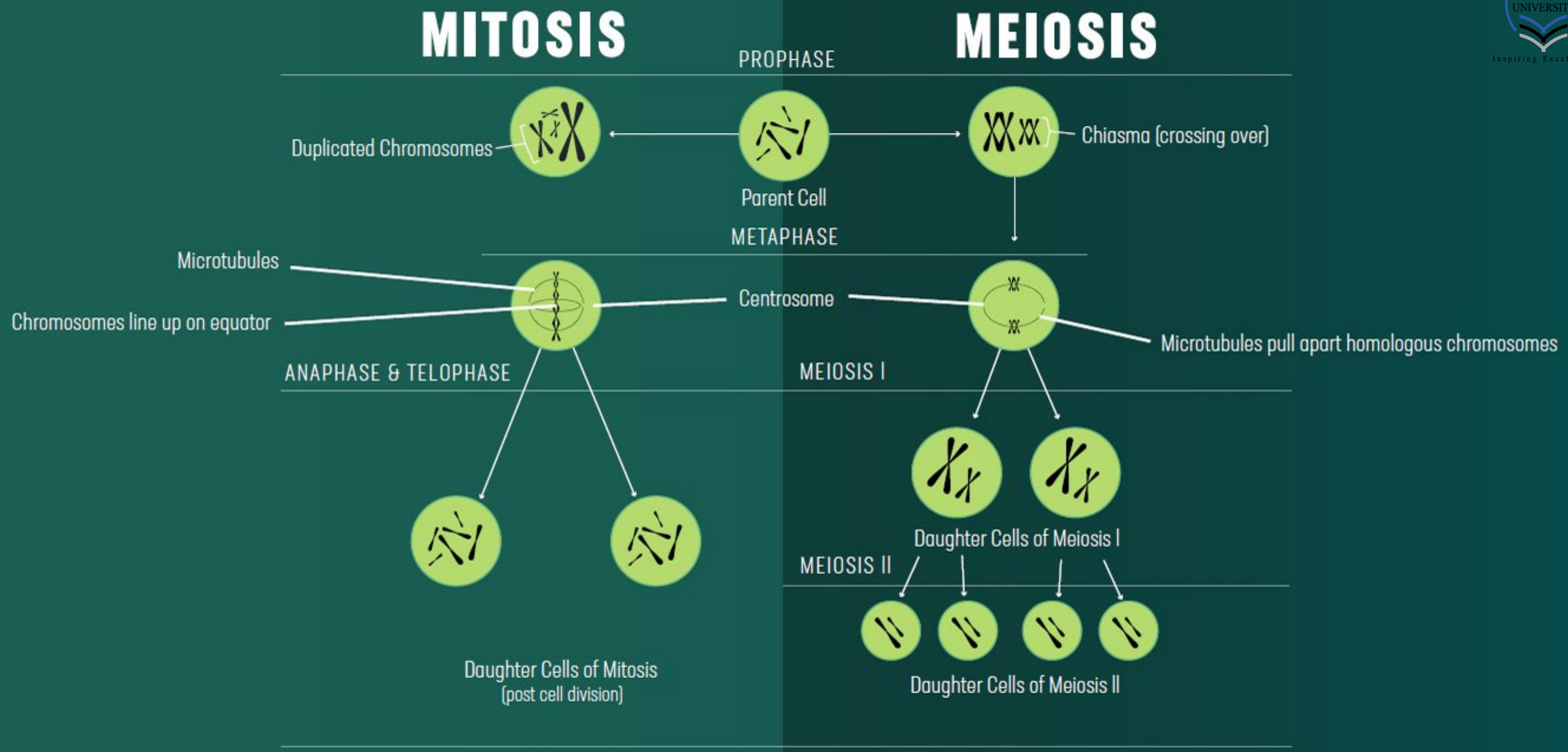
SIGNIFICANCE OF MEIOSIS

1. Gamete production: From the parent cell, gives rise to four haploid cells which are known as gametes -> responsible for sexual reproduction. Depending on the species, meiosis can also create spores.

2. Variation: Due to crossing over in parental chromosomes and random assortment, new genetic combinations occur -> Necessary for evolution.

3. Maintaining chromosome number: The number of chromosomes is halved during meiosis (diploid to haploid); this allows 2 gametes to fuse to form a zygote containing a mixture of paternal and maternal chromosomes -> zygote becomes diploid again

4. Genetic disorder: Malfunction can lead to genetic disorder -> may be passed through generations


MITOSIS
MEIOSIS
PROPHASE
Duplicated Chromosomes
Parent Cell
Chiasma (crossing over)
METAPHASE
Microtubules
Chromosomes line up on equator
Centrosome
Microtubules pull apart homologous chromosomes
ANAPHASE & TELOPHASE
MEIOSIS I
Daughter Cells of Meiosis I
MEIOSIS II
Daughter Cells of Mitosis
(post cell division)
Daughter Cells of Meiosis II

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Mitosis

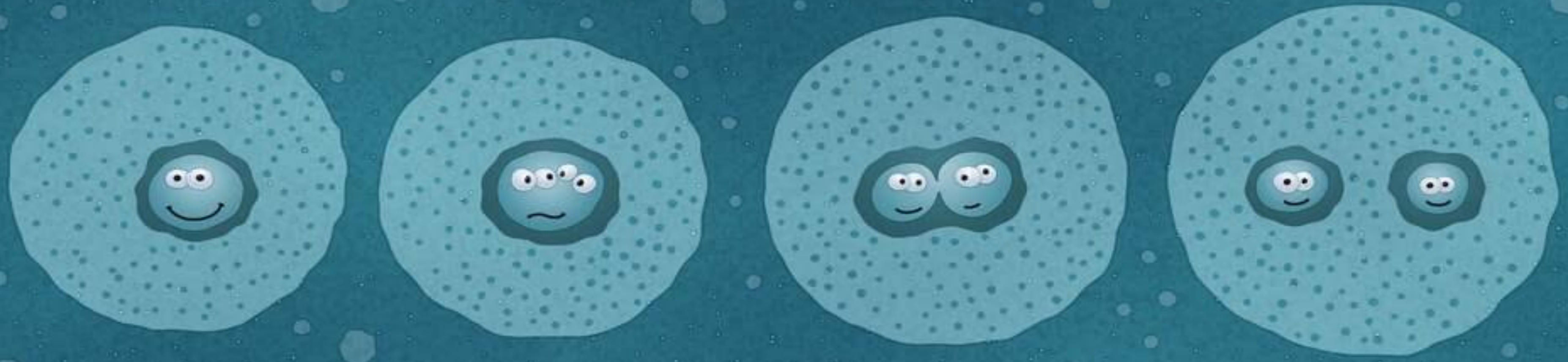
- 4 stages in total (plus interphase)
- Happens in somatic cells
- Purpose is cellular proliferation
- Produces 2 diploid daughter cells
- Chromosome number remains the same
- Genetic variation doesn't change

Same

- Produce new cells
- Similar basic steps
- Start with a single parent cell

Meiosis

- 8 stages in total (plus interphase)
- Happens in germ cells
- Purpose is sexual reproduction
- Produces 4 haploid daughter cells
- Chromosome number is halved in each daughter cell
- Genetic variation increased



THANK YOU